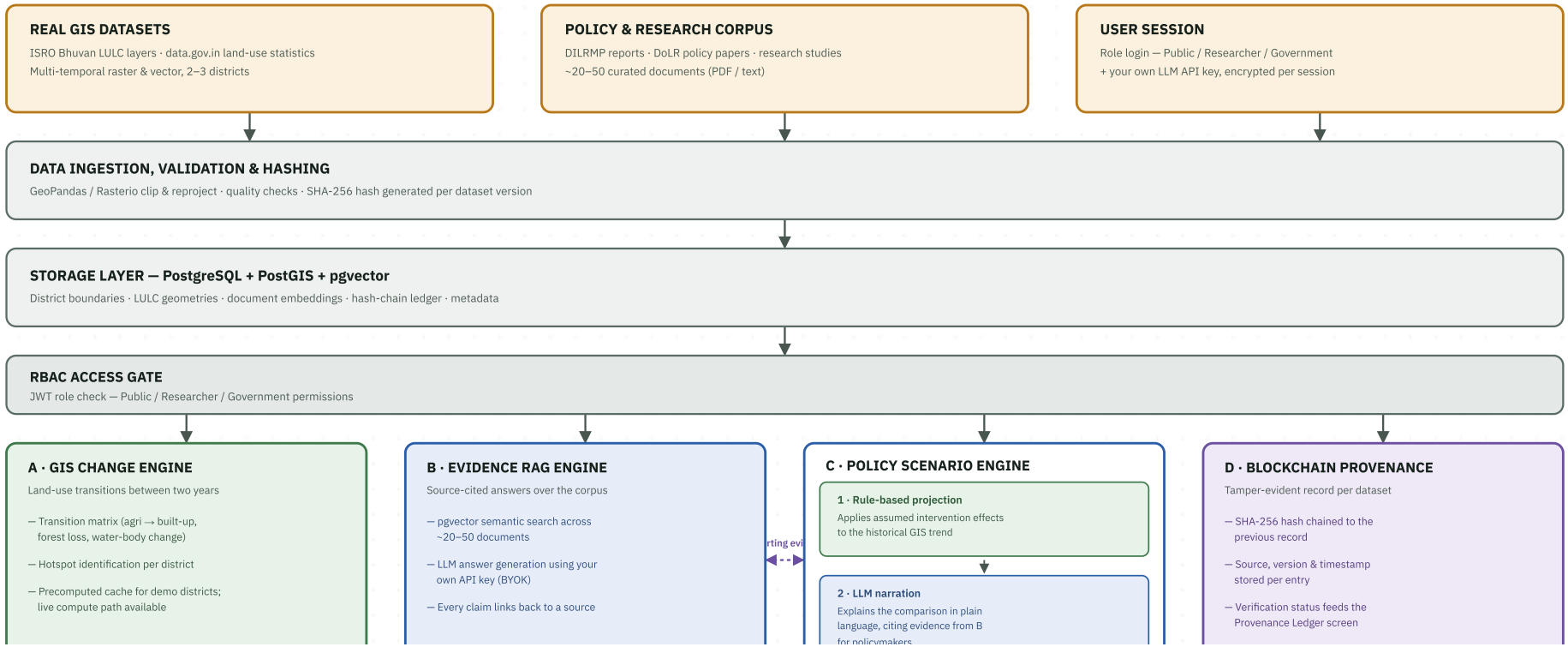
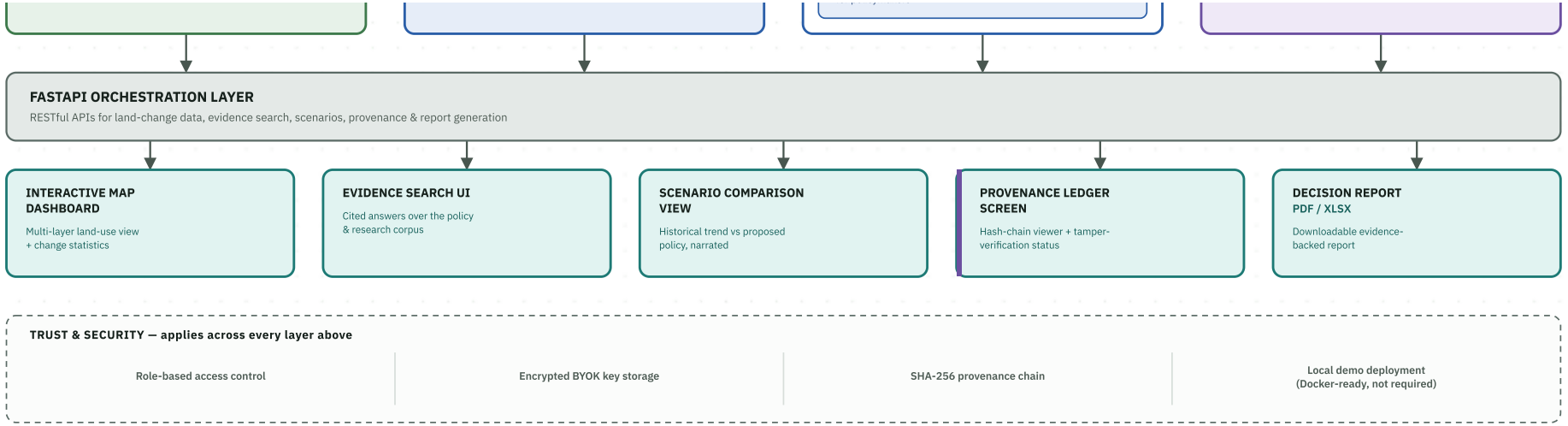
SIH26019 · TEAM PLOTARMOR · MINISTRY OF RURAL DEVELOPMENT — DOLR

BhoomiNITI AI — System Architecture

How real land-governance data becomes an evidence-backed, provenance-verified policy decision — inputs, processing engines, and outputs for the Week 1 MVP.

Input data | Infrastructure | Deterministic processing | AI / LLM reasoning | Blockchain / security | Output





Two real inputs — GIS datasets and the policy corpus — plus a role-and-key user session flow through ingestion, storage and an RBAC gate before reaching four parallel engines. The Policy Scenario Engine draws supporting citations from the Evidence RAG Engine. All four report through one FastAPI layer into five judge-facing outputs, with role-based access and the SHA-256 provenance chain applying across every layer.

Week 1 build notes

- **Provenance** is a simulated SHA-256 hash-chain inside Postgres, not a public testnet — described honestly as a private, testnet-ready ledger, surfaced on its own Provenance Ledger screen.
- **Districts**: 2–3 real districts from Bhuvan / data.gov.in, chosen for the most visually dramatic land-use change.
- **LLM access** is bring-your-own-key, entered in-app and encrypted per session — the same pattern used in the QA-agent and Nova/SemantIQ builds.
- **Change detection** is precomputed and cached for the demo districts; the live GeoPandas/Rasterio path exists but isn't the critical path for the scripted demo.
- **Deployment** is local-only for a team-scripted live demo — Docker is skipped this week, not required.

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